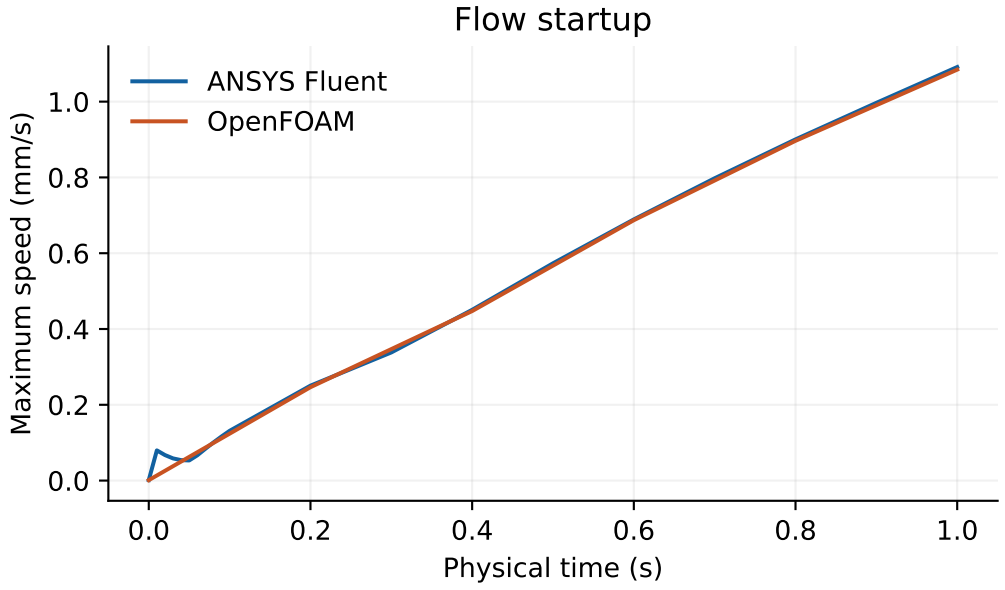
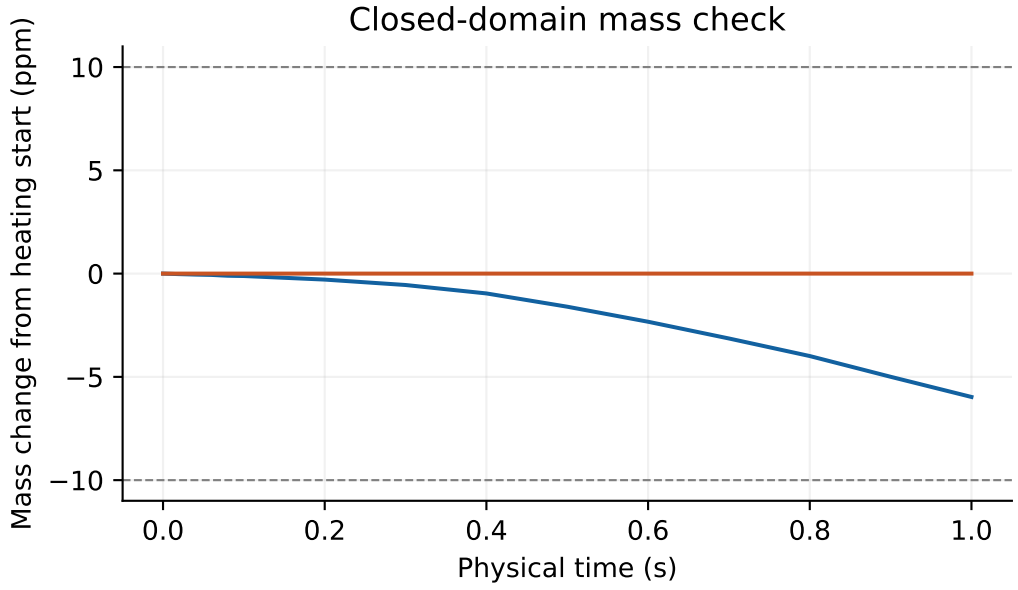
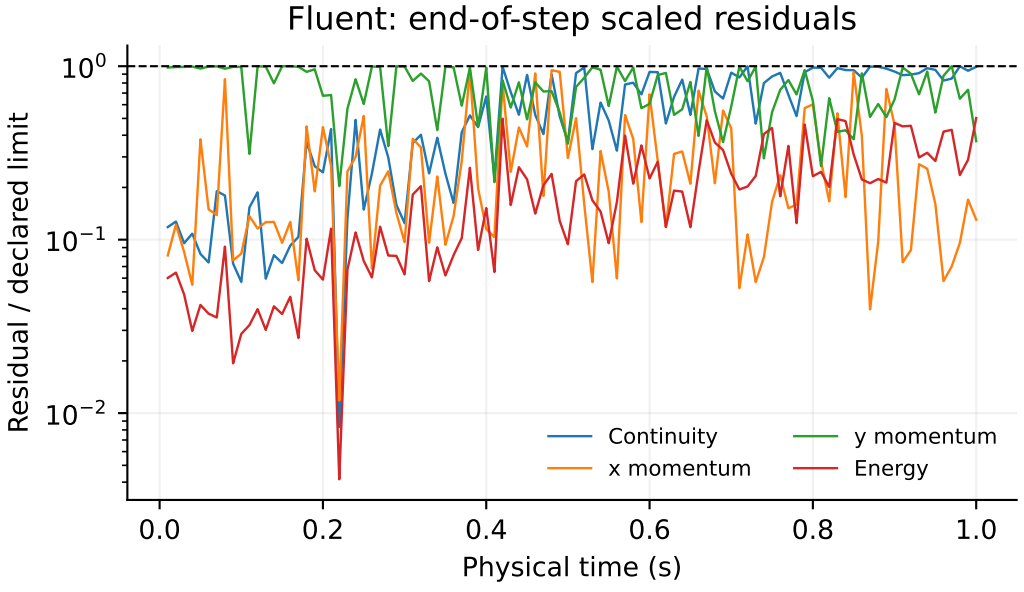
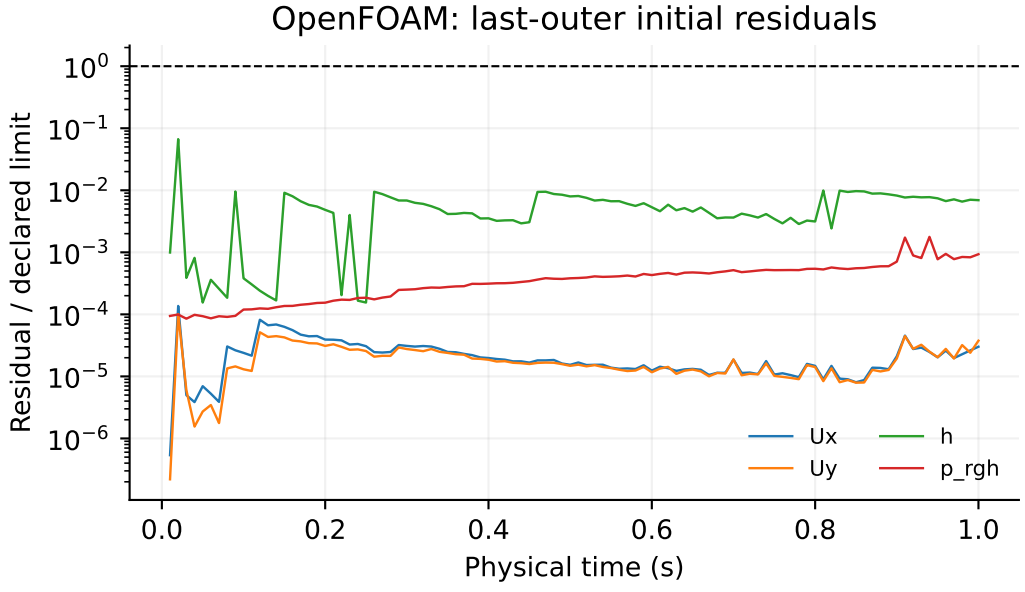
Numerical checks for the saved plume calculation









Residual definitions differ between solvers; these ratios assess each solver against its own limits.
Passing these checks establishes numerical acceptance for this run, not mesh independence or physical validation.